Loglinear Models for Counts

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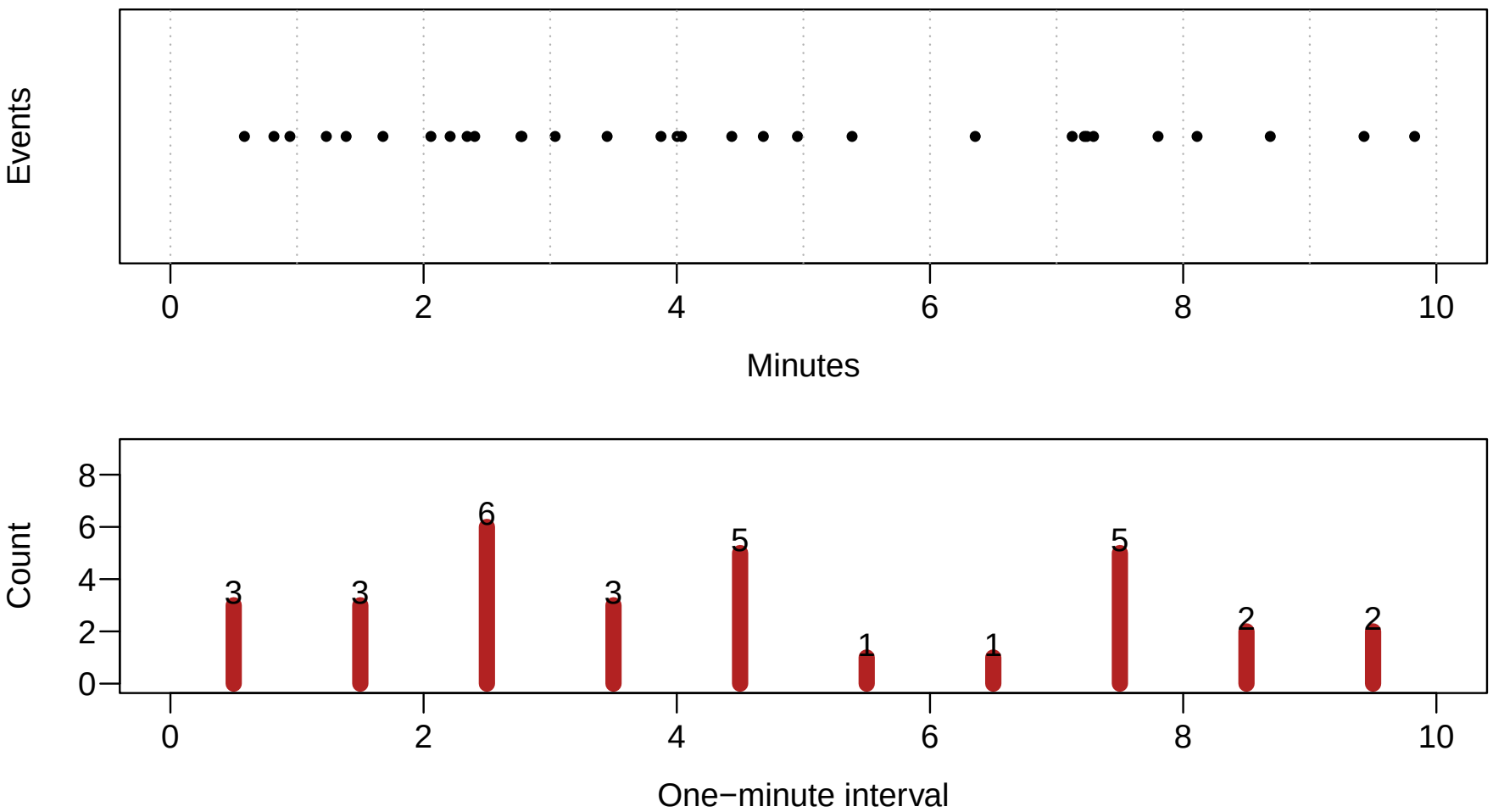
Counting Events

Suppose we count arrivals during a one-minute interval.

- What values can the response take?
- What would it mean for the average count to be 2.5?
- If the underlying rate stays the same, should every minute have the same count?

What would a probability model for this response need to describe?

Events and Counts over Time



Simulated events at a constant rate. What changes if we combine adjacent one-minute intervals?

The Poisson Distribution

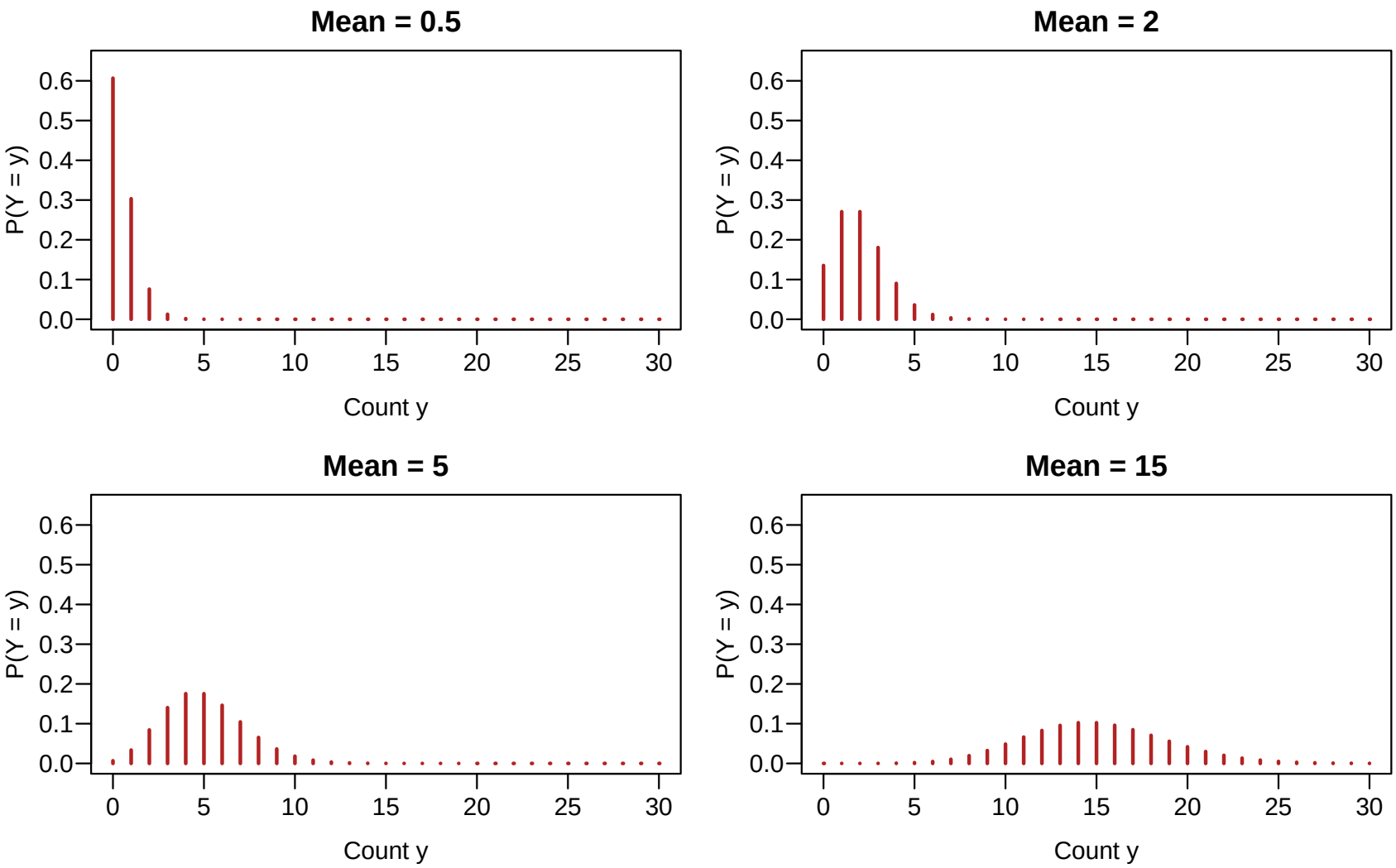
For $Y \sim \text{Poisson}(\mu)$,

$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

- Y is a count, with no fixed upper bound.
- $\mu > 0$ is the expected count for the exposure being observed.
- μ need not be an integer.

If $\mu = 2$, what are $P(Y = 0)$ and $P(Y = 2)$?

Changing the Expected Count



How do the location, spread, chance of zero, and shape change with μ?

Exploring Poisson Probabilities in R

```
> # dpois gives probabilities, ppois cumulative probabilities,
> # qpois quantiles, and rpois random counts. R names the mean "lambda".
> # Start with mu = 2: ask students to predict zeros and P(Y <= 2).
> mu <- 2
> dpois(0:5, lambda = mu)
[1] 0.135335 0.270671 0.270671 0.180447 0.090224 0.036089
> ppois(2, lambda = mu)
[1] 0.67668
> 1 - ppois(2, lambda = mu)          # P(Y > 2), not P(Y >= 2)
[1] 0.32332
```

R uses `lambda` for the mean parameter. What is the difference between `dpois()`, `ppois()`, and `rpois()`?

Mean and Variance

$$E(Y) = \mu, \qquad \text{Var}(Y) = \mu, \qquad \text{SD}(Y) = \sqrt{\mu}.$$

```
> # Repeated observations at the SAME mean. Empirical mean and variance
> # should be close to 4, not exactly 4. A non-integer mean is also
    allowed.
> counts <- rpois(1000, lambda = 4)
> c(mean = mean(counts), variance = var(counts),
+   sd = sd(counts), proportion_zero = mean(counts == 0))
          mean       variance             sd proportion_zero
        3.9440         4.0649         2.0162         0.0170
> c(theory_mean = 4, theory_variance = 4,
+   theory_sd = sqrt(4), theory_zero = exp(-4))
   theory_mean theory_variance       theory_sd    theory_zero
      4.000000       4.000000        2.000000       0.018316
```

If the mean quadruples, what happens to the standard deviation?

Exposure and the Mean Count

At a constant rate λ per minute, the expected count in t minutes is

$$\mu = \lambda t.$$

- Doubling exposure doubles the expected count.
- For independent Poisson counts, adding intervals adds their means.
- A count of 8 in two minutes and a count of 8 in ten minutes suggest different rates.

What assumptions about arrivals might fail just before a class begins?

A Connection to the Binomial

Compare binomial counts with the same mean:

$$\text{Bin}(10, 0.2), \quad \text{Bin}(100, 0.02), \quad \text{Bin}(1000, 0.002).$$

- Many opportunities, each with a small event probability.
- With $np = \mu$ fixed, the binomial approaches Poisson(μ) as n grows and p shrinks.
- The binomial variance $np(1-p)$ approaches μ.

What distinguishes a binomial count from a Poisson count?

When Would Poisson Be Reasonable?

- Arrivals in equal intervals at a stable rate.
- Students entering a classroom just before class.
- Counts pooled from locations with very different underlying rates.

Which situations might have more variability than a simple Poisson model allows? Why?

Mean = variance applies at a fixed modeled mean. It need not hold after pooling observations with different means.

Modeling Counts in a Contingency Table

Previously, we modeled an individual's response probability. Now each **cell count** is a response:

$$N_{ij} \sim \text{Poisson}(\mu_{ij}).$$

- X and Y identify the row and column categories.
- N_{ij} is the observed count in cell (i, j).
- μ_{ij} is that cell's expected count.

The classification variables are treated symmetrically. Neither must be designated as the outcome variable.

Independence and Expected Counts

Under independence,

$$\pi_{ij} = \pi_{i+}\pi_{+j}.$$

For a table with total n,

$$\mu_{ij} = n\pi_{ij} = n\pi_{i+}\pi_{+j}.$$

Taking logs gives an additive relationship:

$$\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}.$$

Which parts depend on the row? The column? Both?

The Loglinear Independence Model

We write the model as

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y.$$

- Poisson random component and log link.
- Row and column effects add on the log-mean scale.
- Effects multiply on the expected-count scale.
- No row-by-column interaction: independence of X and Y.

The superscripts X and Y label variables; they are not powers.

Happiness and Belief in Heaven

General Social Survey respondents reported happiness and whether they believed in heaven.

```
> HappyHeaven <- read.table("data/HappyHeaven.dat", header = TRUE)
> HappyHeaven$happy <- factor(HappyHeaven$happy,
+                             levels = c("not", "pretty", "very"))
> HappyHeaven$heaven <- factor(HappyHeaven$heaven,
+                              levels = c("no", "yes"))
> xtabs(count ~ happy + heaven, data = HappyHeaven)
        heaven
happy     no yes
  not     32 190
  pretty 113 611
  very    51 326
```

Three happiness categories and two belief categories give six cell counts to model.

The Independence Model in R

```
> fit <- glm(count ~ happy + heaven, family = poisson,
+            data = HappyHeaven)
> summary(fit)
Call:
glm(formula = count ~ happy + heaven, family = poisson, data =
    HappyHeaven)
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)    3.4931     0.0941   37.13  < 2e-16 ***
happypretty    1.1821     0.0767   15.41  < 2e-16 ***
happyvery      0.5296     0.0846    6.26  3.9e-10 ***
heavenyes      1.7492     0.0774   22.60  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for poisson family taken to be 1)
    Null deviance: 1019.87238  on 5  degrees of freedom
Residual deviance:    0.89111  on 2  degrees of freedom
AIC: 49.5
Number of Fisher Scoring iterations: 3
```

Observed and Fitted Counts

```
> HappyHeaven$fitted <- fitted(fit)
> HappyHeaven
   happy heaven count  fitted
1    not     no    32  32.889
2    not    yes   190 189.111
3 pretty     no   113 107.259
4 pretty    yes   611 616.741
5   very     no    51  55.852
6   very    yes   326 321.148
```

Do the fitted counts look familiar from chi-squared tests of independence?

The Familiar Independence Fit

$$\hat{\mu}_{ij} = \frac{n_{i+} n_{+j}}{n}.$$

```
> # Check that the fitted counts are exactly the familiar independence
    counts.
> tab <- xtabs(count ~ happy + heaven, data = HappyHeaven)
> expected <- outer(rowSums(tab), colSums(tab)) / sum(tab)
> round(expected, 2)
           no    yes
not     32.89 189.11
pretty 107.26 616.74
very    55.85 321.15
> c(G2 = deviance(fit), df = df.residual(fit),
+   p = pchisq(deviance(fit), df.residual(fit), lower.tail = FALSE))
     G2      df       p
0.89111 2.00000 0.64047
```

Six cells minus four parameters gives two residual degrees of freedom. The model shows no evidence of lack of fit.

Interpreting the Coefficients

The fitted model has reference categories `not` (not too happy) and `no` belief in heaven.

- $e^{3.4931} \approx 32.89$: expected count in the reference cell.
- $e^{1.1821} \approx 3.26$: pretty/not expected-count ratio within either heaven category.
- $e^{1.7492} \approx 5.75$: yes/no expected-count ratio within each happiness category.

The last ratio is the odds of belief in heaven in each row. The model assumes those odds are the same across happiness levels.

Association as an Interaction

Add row-by-column terms:

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}.$$

- The column effect may now depend on the row.
- There are $(r-1)(c-1)$ additional parameters.
- Total parameters: $1 + (r-1) + (c-1) + (r-1)(c-1) = rc$.

This is the saturated model: one fitted mean per cell.

The Saturated Model in R

```
> fit_sat <- glm(count ~ happy * heaven, family = poisson,
+                data = HappyHeaven)
> summary(fit_sat)
Call:
glm(formula = count ~ happy * heaven, family = poisson, data =
    HappyHeaven)
Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)             3.4657     0.1768   19.61   <2e-16 ***
happypretty             1.2617     0.2002    6.30    3e-10 ***
happyvery               0.4661     0.2255    2.07    0.039 *
heavenyes               1.7813     0.1911    9.32   <2e-16 ***
happypretty:heavenyes  -0.0936     0.2168   -0.43    0.666
happyvery:heavenyes     0.0738     0.2433    0.30    0.762
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for poisson family taken to be 1)
    Null deviance: 1.0199e+03  on 5  degrees of freedom
Residual deviance: 2.2204e-16  on 0  degrees of freedom
AIC: 52.61
Number of Fisher Scoring iterations: 2
```

Interactions and Odds Ratios

For a 2×2 table,

$$\log \theta = \log \mu_{11} + \log \mu_{22} - \log \mu_{12} - \log \mu_{21}.$$

Substituting the loglinear model cancels the intercept and main effects:

$$\log \theta = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}.$$

With reference-category coding, only λ_{22}^{XY} remains. Exponentiating the interaction coefficient gives the odds ratio.

Happiness–Heaven Odds Ratios

```
> # Under the stated reference categories, these interactions
    are log odds ratios.
> exp(coef(fit_sat)[c("happypretty:heavenyes",
+                     "happyvery:heavenyes")])
happypretty:heavenyes   happyvery:heavenyes
              0.91067                1.07657
```

- Pretty versus not: estimated belief odds ratio ≈ 0.91.
- Very versus not: estimated belief odds ratio ≈ 1.08.

These are comparisons of belief odds between happiness categories, rather than the yes/no count ratio within a row.

Review: Two-Way Tables

1. What is the response in a loglinear model?
2. Why does independence imply an additive log-mean model?
3. Why can a main effect be highly significant even when the variables are independent?
4. What restriction disappears when we add an interaction?

Three-Way Tables

For expected cell counts μ_{ijk}, possible terms include

- Main effects: X, Y, Z.
- Pairwise interactions: XY, XZ, YZ.
- Three-way interaction: XYZ.

Highest-order terms	Model
(X, Y, Z)	Mutual independence
(XZ, YZ)	$X \perp Y \mid Z$
(XY, XZ, YZ)	Homogeneous association
(XYZ)	Saturated

Conditional Independence

Consider

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}.$$

- XZ and YZ associations are allowed.
- There is no XY or XYZ term.
- Within each level of Z, the conditional XY odds ratio is 1.

Thus $X \perp Y \mid Z$. Does this require marginal independence of X and Y?

Homogeneous Association

Include all pairwise interactions, with no three-way interaction:

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}.$$

- Pairwise conditional odds ratios may differ from 1.
- The XY odds ratio is the same at every level of Z.
- Similarly, the XZ and YZ conditional odds ratios are constant across the remaining variable.

Adding XYZ allows the conditional associations to vary across strata.

Student Substance Use

A = alcohol, C = cigarettes, M = marijuana. Each records whether a high-school student had ever used the substance.

```
> Drugs <- read.table("data/Substance.dat", header = TRUE)
> # Use no as reference throughout, so interaction coefficients match
    the text.
> Drugs$A <- factor(Drugs$alcohol, levels = c("no", "yes"))
> Drugs$C <- factor(Drugs$cigarettes, levels = c("no", "yes"))
> Drugs$M <- factor(Drugs$marijuana, levels = c("no", "yes"))
> ftable(xtabs(count ~ A + C + M, data = Drugs))
         M  no yes
A   C
no  no    279   2
    yes    43   3
yes no    456  44
    yes   538 911
```

Fitting Three-Way Models

```
> mutual <- glm(count ~ A + C + M, family = poisson, data = Drugs)
> conditional <- glm(count ~ A * M + C * M,
+                    family = poisson, data = Drugs)
> homogeneous <- glm(count ~ (A + C + M)^2,
+                    family = poisson, data = Drugs)
> saturated <- glm(count ~ A * C * M,
+                  family = poisson, data = Drugs)
```

In an R formula, `(A + C + M)^2` includes all main effects and pairwise interactions. `A * C * M` also includes the three-way interaction.

The Homogeneous Association Model in R

```
> summary(homogeneous)
Call:
glm(formula = count ~ (A + C + M)^2, family = poisson, data = Drugs)
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   5.6334     0.0597   94.36  < 2e-16 ***
Ayes          0.4877     0.0758    6.44  1.2e-10 ***
Cyes         -1.8867     0.1627  -11.60  < 2e-16 ***
Myes         -5.3090     0.4752  -11.17  < 2e-16 ***
Ayes:Cyes     2.0545     0.1741   11.80  < 2e-16 ***
Ayes:Myes     2.9860     0.4647    6.43  1.3e-10 ***
Cyes:Myes     2.8479     0.1638   17.38  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for poisson family taken to be 1)
    Null deviance: 2851.46098  on 7  degrees of freedom
Residual deviance:    0.37399  on 1  degrees of freedom
AIC: 63.42
Number of Fisher Scoring iterations: 4
```

Conditional Odds Ratios

```
> exp(coef(homogeneous)[c("Ayes:Cyes", "Ayes:Myes",
    "Cyes:Myes")])
Ayes:Cyes Ayes:Myes Cyes:Myes
   7.8032   19.8066   17.2513
```

- *AC*: about 7.8, at either level of marijuana use.
- *AM*: about 19.8, at either level of cigarette use.
- *CM*: about 17.3, at either level of alcohol use.

The common conditional odds ratios need not equal the marginal odds ratios.

Fitted Counts under Different Models

```
> # Reproduce Table 7.2. Use the file's row order, which matches the
    text.
> round(data.frame(observed = Drugs$count,
+   mutual = fitted(mutual), conditional = fitted(conditional),
+   homogeneous = fitted(homogeneous), saturated = fitted(saturated)),
    2)
  observed mutual conditional homogeneous saturated
1      911 539.98      909.24      910.38       911
2      538 740.23      438.84      538.62       538
3       44 282.09       45.76       44.62        44
4      456 386.70      555.16      455.38       456
5        3  90.60        4.76        3.62         3
6       43 124.19      142.16       42.38        43
7        2  47.33        0.24        1.38         2
8      279  64.88      179.84      279.62       279
```

Rows follow the original data-file order. The homogeneous association model is close to the observed table; the simpler models miss important features.

Conditional and Marginal Association

Under (AM, CM), alcohol and cigarettes are independent given marijuana use.

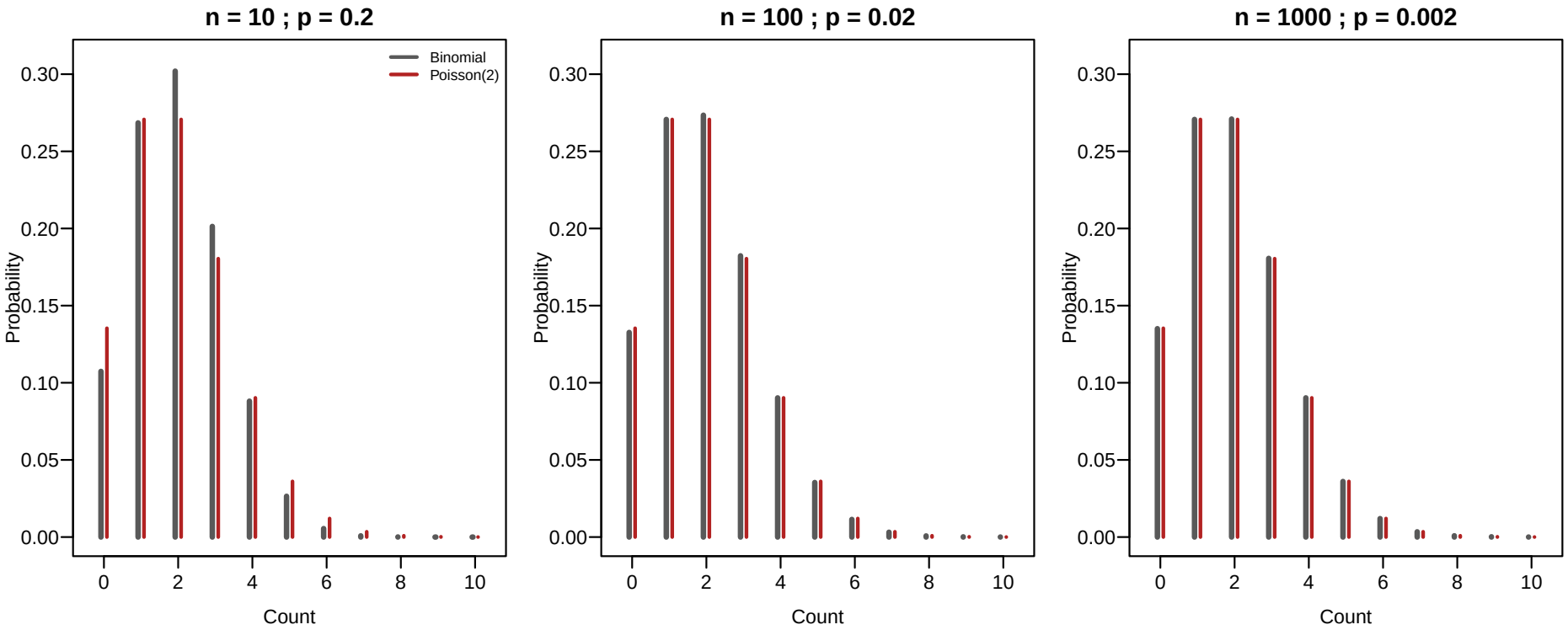
```
> c(no_marijuana = OR(ci_table[, , "no"]),
+   yes_marijuana = OR(ci_table[, , "yes"]),
+   marginal = OR(apply(ci_table, c(1, 2), sum)))
 no_marijuana yes_marijuana      marginal
       1.0000        1.0000        2.7497
```

These odds ratios use *fitted counts from the conditional-independence model*. Both conditional odds ratios are 1, but the marginal odds ratio is about 2.75. Conditional independence does not imply marginal independence.

Review: Three-Way Tables

1. Which terms allow conditional association between two variables?
2. What does omitting the three-way interaction assume?
3. How does conditional independence differ from homogeneous association?
4. What would a saturated model allow that the homogeneous association model does not?

Appendix: Binomial Counts Approaching Poisson



The means stay at 2 as the number of opportunities grows and the event probability shrinks.

Appendix: Fixed and Random Totals

Independent Poisson cell counts have a random total:

$$N_{ij} \sim \text{Poisson}(\mu_{ij}), \qquad N = \sum_{ij} N_{ij}.$$

Conditional on $N = n$, the counts have a multinomial distribution with

$$\pi_{ij} = \frac{\mu_{ij}}{\sum_{ab} \mu_{ab}}.$$

- Fixed-total multinomial counts are dependent.
- For the models used here, the Poisson fit with an intercept gives the same fitted counts as the corresponding multinomial model.

Sources

- Agresti, *An Introduction to Categorical Data Analysis*, third edition, Section 7.1, pp. 194–200.
- Happiness and belief in heaven: Section 7.1.3.
- Student substance use: Tables 7.1–7.3, Section 7.1.7.
- Original data: Alan Agresti's public data repository.

The opening Poisson-process and count simulations are constructed lecture examples. All displayed R results and probability plots come from the companion script.